



JABchem



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Past Papers Higher Chemistry

2017 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	90+	75.0%	30.3%
B	79+	65.8%	24.2%
C	68+	56.7%	21.3%
D	62+	51.7%	8.4%
No award	<62	<51.7%	15.8%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.2 /20	49.1 /80	13.9 /20

2017 Higher Chemistry Marking Scheme

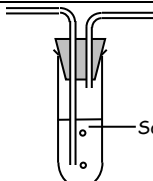
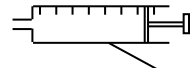
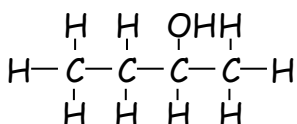
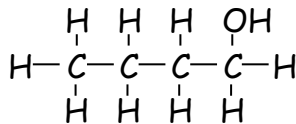
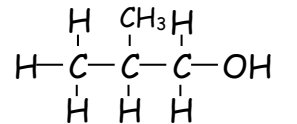
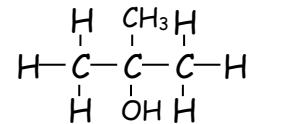
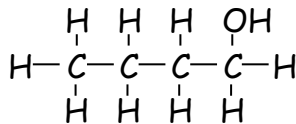
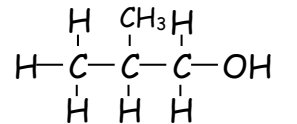
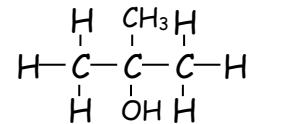
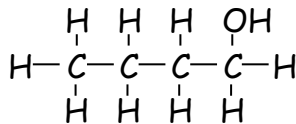
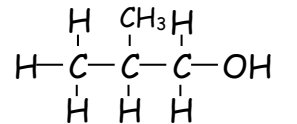
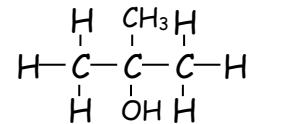
MC Qu	Answer	% Pupils Correct	Reasoning
1	A	81	☒ A Electronegativity difference C-I = 2.6-2.5 = 0.1 ∴ least polar bond ☒ B Electronegativity difference C-I = 4.0-2.5 = 1.5 ∴ most polar bond ☒ C Electronegativity difference C-I = 3.0-2.5 = 0.5 ☒ D Electronegativity difference C-I = 2.8-2.5 = 0.3
2	D	86	☒ A non-polar hydrocarbon ∴ molecule is non-polar ∴ molecule is insoluble in water ☒ B one -OH groups ∴ molecule is polar ∴ molecule is soluble in water ☒ C aldehyde group ∴ molecule is <i>slightly</i> polar ∴ molecule is slightly soluble in water ☒ D three -OH groups ∴ molecule is polar ∴ molecule is very soluble in water
3	C	76	☒ A Electronegativity of sulphur = 2.5 ☒ B Electronegativity of silicon = 1.9 ∴ least attraction for electrons ☒ C Electronegativity of nitrogen = 3.0 ∴ greatest attraction for electrons ☒ D Electronegativity of hydrogen = 2.2
4	B	86	☒ A Covalent networks have high melting points e.g. B, C (diamond), C (graphite), Si ☒ B Phosphorus P ₄ is a covalent molecular substance also S ₈ and C ₆₀ (fullerene) ☒ C monatomic substances are found in group 0 e.g. He, Ne, Ar ☒ D metallic lattices are found in metal elements e.g. Li, Be, Na, Mg, Al, K, Ca
5	D	82	☒ A water H ₂ O contains polar -OH groups which are attracted to charged rod ☒ B propanone contains polar C=O group which is attracted to charged rod ☒ C propanol C ₃ H ₇ OH contains polar -OH group which is attracted to charged rod ☒ D hexane C ₆ H ₁₄ is non-polar which is not attracted to charged rod
6	C	94	$2P_2H_4 + 7O_2 \longrightarrow P_4O_{10} + 4H_2O$
7	B	95	☒ A longest chain containing -OH functional group is three carbons long ☒ B longest chain = 3, -OH on C ₁ and 2x -CH ₃ methyl groups on C ₂ ☒ C functional group located on C ₁ to give functional group lowest numbering system ☒ D longest chain containing -OH function group is three carbons not five carbons
8	C	65	☒ A C ₁₅ H ₂₉ COOH is unsaturated ∴ C ₁₅ H ₂₉ COOH contains one C=C double bond ☒ B C ₁₅ H ₃₁ COOH is saturated ∴ C ₁₅ H ₃₁ COOH contains no C=C double bonds ☒ C C ₁₇ H ₃₁ COOH is unsaturated ∴ C ₁₇ H ₃₁ COOH contains two C=C double bonds ☒ D C ₁₇ H ₃₅ COOH is saturated ∴ C ₁₇ H ₃₅ COOH contains no C=C double bonds
9	A	98	☒ A Activation is not a step in a free radical chain reaction mechanism. ☒ B Initiation: Free-radicals are produced e.g. Cl ₂ → 2Cl• ☒ C Propagation: Free-radicals on both sides of arrow e.g. CH ₄ + Cl• → CH ₃ • + HCl ☒ D Termination: Free-radicals join together e.g. CH ₃ • + Cl• → CH ₃ Cl
10	C	68	☒ A methylpropanoate C ₃ H ₆ O ₂ is not an isomer of ethyl propanoate C ₅ H ₁₀ O ₂ ☒ B pentan-2-one C ₅ H ₁₀ O is not an isomer of ethyl propanoate C ₅ H ₁₀ O ₂ ☒ C pentanoic acid C ₅ H ₁₀ O ₂ is an isomer of ethyl propanoate C ₅ H ₁₀ O ₂ ☒ D pentane-1,2-diol C ₅ H ₁₂ O ₂ is not an isomer of ethyl propanoate C ₅ H ₁₀ O ₂
11	B	61	☒ A essential oils are volatile compounds to allow release of aroma ☒ B essential oils are volatile compounds which do not dissolve in water. ☒ C essential oils are non-water soluble ☒ D essential oils are non-water soluble
12	C	85	☒ A no. of moles of water formed dependent on formula of hydrocarbon ☒ B no. of moles of carbon dioxide formed dependent on formula of hydrocarbon ☒ C one mole of hydrocarbon burned completely in plentiful supply of oxygen ☒ D enthalpy of combustion has a plentiful supply of oxygen

13	B	67	☒ A Fluorine is a strong oxidising agent as it is located on bottom left of ECS ☒ B Lithium is a strong reducing agent as it is located on top right of ECS ☒ C Calcium is a reducing agent as it is located towards top right of ECS ☒ D Iodine is a oxidising agent as it is located towards bottom left of ECS																												
14	B	79	$\text{atom economy} = \frac{\text{mass of useful products}}{\text{total mass of reactants}} \times 100 = \frac{47.9}{189.9 + (2 \times 24.3)} \times 100$																												
15	C	84	The most reproducible set of results are Student C as they have three results within $\pm 0.2 \text{ cm}^3$.																												
16	A	83	$ \begin{array}{c} \text{H} & \text{O} & \text{H} & \text{H} & \text{H} \\ & & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \text{CH}_3 & \\ \text{H} & & \text{H} & & \text{H} \end{array} + \text{HCN} \longrightarrow \begin{array}{c} \text{H} & \text{OH} & \text{H} & \text{H} & \text{H} \\ & & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & & & \\ \text{H} & \text{CN} & \text{H} & \text{CH}_3 & \text{H} \end{array} $																												
17	D	89	☒ A All reactions have an enthalpy change as bonds break and new bonds are formed ☒ B concentrations of reactants and products are not equal at equilibrium ☒ C activation energy of forward/reverse reactions only equal If $\Delta H = 0$ ☒ D at equilibrium, rate of forward reaction equals rate of reverse reaction																												
18	A	76	<table border="1"> <thead> <tr> <th colspan="2">Bond Breaking Steps</th> <th colspan="2">Bond Forming Steps</th> </tr> </thead> <tbody> <tr> <td>1x H-H</td> <td>= 1x436kJ = 436kJ</td> <td>2x H-Br</td> <td>= 2x366kJ = 732kJ</td> </tr> <tr> <td>1x Br-Br</td> <td>= 1x194kJ = 194kJ</td> <td></td> <td></td> </tr> <tr> <td colspan="2">Total = 630kJ</td> <td colspan="2">Total = 732kJ</td> </tr> <tr> <td>ΔH</td> <td>= ΣBond enthalpies for bonds broken</td> <td>-</td> <td>ΣBond enthalpies for bonds formed</td> </tr> <tr> <td>ΔH</td> <td>= 630</td> <td>-</td> <td>732</td> </tr> <tr> <td>ΔH</td> <td>= -102 kJ mol⁻¹</td> <td></td> <td></td> </tr> </tbody> </table>	Bond Breaking Steps		Bond Forming Steps		1x H-H	= 1x436kJ = 436kJ	2x H-Br	= 2x366kJ = 732kJ	1x Br-Br	= 1x194kJ = 194kJ			Total = 630kJ		Total = 732kJ		ΔH	= Σ Bond enthalpies for bonds broken	-	Σ Bond enthalpies for bonds formed	ΔH	= 630	-	732	ΔH	= -102 kJ mol ⁻¹		
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19	D	93	☒ A glycerol has three -OH hydroxyl groups ☒ B glycerol has three carbon atoms in compound ☒ C glycerol has three -OH hydroxyl groups but no -COOH carboxyl group ☒ D glycerol has three carbons and three -OH hydroxyl groups																												
20	B	70	☒ A catalyst increases rate of the forward reaction (and the reverse reaction too) ☒ B catalyst increases rate of forward reaction but no change in position of equilibrium ☒ C catalysts do no change position of equilibrium but equilibrium is achieved faster ☒ D catalysts do no change position of equilibrium but equilibrium is achieved faster																												

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Long Qu	Answer	Reasoning																
1a	Si	<table><tr><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P₄</td><td>S₈</td><td>Cl₂</td><td>Ar</td></tr><tr><td>metallic bonding</td><td>metallic bonding</td><td>metallic bonding</td><td>covalent network</td><td>covalent molecular</td><td>covalent molecular</td><td>covalent molecular</td><td>monatomic</td></tr></table>	Na	Mg	Al	Si	P ₄	S ₈	Cl ₂	Ar	metallic bonding	metallic bonding	metallic bonding	covalent network	covalent molecular	covalent molecular	covalent molecular	monatomic
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metallic bonding	metallic bonding	metallic bonding	covalent network	covalent molecular	covalent molecular	covalent molecular	monatomic											
1b(i)	Electrons closer to nucleus across period	Atoms decrease in size across period due to increased nuclear charge. Electrons are harder to remove if they are closer to the nucleus																
1b(ii)	$\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$	<u>1st Ionisation Energy:</u> $\text{Mg}(\text{g}) \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$ 1 mole of electrons removed from one mole of atoms in the gaseous state <u>2nd Ionisation Energy:</u> $\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$ 1 mole of electrons removed from one mole of 1+ ions in the gaseous state																
1b(iii)	3 rd ionisation is low as full outer shell is created. 4 th Ionisation is higher as it breaks full outer shell	<table><tr><td>Third ionisation energy creates a full outer shell.</td><td>$\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^-$ 2,8,1 2,8</td></tr><tr><td>Fourth ionisation energy breaks into a full outer shell.</td><td>$\text{Al}^{3+}(\text{g}) \rightarrow \text{Al}^{4+}(\text{g}) + \text{e}^-$ 2,8 2,7</td></tr></table>	Third ionisation energy creates a full outer shell.	$\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^-$ 2,8,1 2,8	Fourth ionisation energy breaks into a full outer shell.	$\text{Al}^{3+}(\text{g}) \rightarrow \text{Al}^{4+}(\text{g}) + \text{e}^-$ 2,8 2,7												
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1c	Answer to include:	Argon is a monatomic gas in Group 0 which has only single atoms in its structure. The only type of Van der Waals forces between atoms of argon are London dispersion forces. London dispersion forces are the weakest form of intermolecular forces and this results in a low boiling point. Chlorine is a diatomic molecule but is non-polar due to both atoms having the same electronegativity. The only type of intermolecular forces between chlorine atoms are London dispersion forces. Chlorine has a higher melting point than argon because as it is diatomic and this increases the number of electrons in chlorine molecules compared to argon atoms. The increased number of electrons make a temporary dipole more likely and increases the strength of London dispersion forces, bringing molecules of Cl ₂ closer together than atoms of Ar and raises the boiling point of Cl ₂ higher than Ar.																
2a(i)	Answer to include:	436kJ of energy required to break one mole of H-H bonds 151 kJ of energy required to break one mole of I-I bonds I-I bonds much more likely to break than H-H bonds																
2a(ii)	Diagram showing:																	
2a(iii) Part A	Equilibrium shifts to left	In increase in temperature would favour the endothermic reaction The endothermic reaction is the reverse reaction so equilibrium shifts to left as products are turned into reactants.																
2a(iii) Part B	Equal number of moles of gas on each side of equation	$\begin{array}{ccc} \text{H}_2(\text{g}) & + & \text{I}_2(\text{g}) & \rightleftharpoons & 2\text{HI}(\text{g}) \\ 1\text{mol} & & 1\text{mol} & & 2\text{mol} \\ \underbrace{\hspace{1.5cm}} & & & & \\ & & 2\text{mol} & & 2\text{mol} \end{array}$																

2a(iv) Part A	Activated Complex	Activated Complex is the unstable arrangement of atoms when the reactants bonds are half broken and the product bonds are half formed.						
2a(iv) Part B	-9.6	Enthalpy change = 173.2kJ - 182.8kJ = -9.6kJ mol ⁻¹						
2a(iv) Part C	Decreases activation energy	Catalysts decrease the activation energy of both the forward and reverse reactions						
2b(i)	To prevent dilution of reactants	If any water remains in the beaker prior to transfer of reactants the reactant concentration would decrease.						
2b(ii)	122.1	$\text{rate} = \frac{1}{\text{time}} \therefore \text{time} = \frac{1}{\text{rate}} = \frac{1}{0.00819 \text{ s}^{-1}} = 122.1 \text{ s}$						
2b(iii)	Decreases number of collisions at lower concentration	Lower concentration of Iodide I ⁻ ions results in less collisions between I ⁻ ions and the persulphate S ₂ O ₈ ²⁻ ions. Successful collisions which leads to products being formed requires both the correct collision geometry and collision energy to occur.						
3	Open Question Answer to Include:	<table border="1"> <thead> <tr> <th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr> </thead> <tbody> <tr> <td>Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr> </tbody> </table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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4a(i)	pentan-1-ol	hydrolysis						
4a(ii)	Diagram showing:	Sodium hydroxide reacts with butanoic acid to form sodium butanoate $\text{Na}^+ \quad ^-\text{O}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{H}$						
4b(i)	esters	Fats and oils are esters formed by the condensation reactions between glycerol and three fatty acids						
4b(ii)	Soaps	Salts of fatty acids from the hydrolysis of fats/oils have and polar head and a non-polar tail which allows use as a soap.						
4b(iii)	394.3 kJ	$\begin{aligned} &\text{C}_{17}\text{H}_{35}\text{COOH} + 26\text{O}_2 \rightarrow 18\text{CO}_2 + 18\text{H}_2\text{O} \\ &1\text{mol} \qquad\qquad\qquad 18\text{mol} \\ &1\text{mol} \times \frac{1}{18} \qquad\qquad\qquad 1\text{mol} \\ &= 0.0556\text{mol} \\ &\text{mass} = \text{no. of mol} \times \text{gfm} \\ &= 0.0556\text{mol} \times 284 \text{ g mol}^{-1} \\ &= 15.8\text{g} \end{aligned}$ $\begin{aligned} 15.8\text{g stearic acid} &\longleftrightarrow 623\text{kJ} \\ 10\text{g stearic acid} &\longleftrightarrow 623\text{kJ} \times \frac{10}{15.8} \\ &= 394.3\text{kJ} \end{aligned}$						

5a(i)	Answer to include:	 e.g. bubbling gases through sodium hydroxide solution	 e.g. syringe to collect gas or downwards displacement of air																
5a(ii)	Answer showing:	$\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.40\text{g}}{126.1 \text{ g mol}^{-1}} = 0.00317\text{mol}$ $\text{Na}_2\text{SO}_3 + 2\text{HCl} \longrightarrow \text{H}_2\text{O} + 2\text{NaCl} + \text{SO}_2$ 1mol 0.00317mol 2mol 0.00634mol (needed) $n \text{ HCl} = v \times c = 0.05\text{litres} \times 1.0 \text{ mol l}^{-1} = 0.05\text{mol (available)}$ There is more HCl available for the reaction than is needed to completely react with all Na₂SO₃ ∴ Na₂SO₃ is the limiting reactant.																	
5b	-1075	① C + O₂ → CO₂ ΔH= -393.5 kJ mol⁻¹ ② S + O₂ → SO₂ ΔH= -296.8 kJ mol⁻¹ ③ C + 2S → CS₂ ΔH= +87.9 kJ mol⁻¹ ① C + O₂ → CO₂ ΔH= -393.5 kJ ②x2 2S + 2O₂ → 2SO₂ ΔH= -593.6 kJ ③x-1 CS₂ → C + 2S ΔH= -87.9 kJ Add CS₂ + 3O₂ → CO₂ + 2SO₂ ΔH= -1075 kJ mol⁻¹ ①+②'+③'																	
5c(i)	165 g l ⁻¹	Join line of best fit ignoring the rogue result. Extrapolate up from 10°C to line and read solubility from graph																	
5c(ii)	Answer to include:	CO ₂ is a non-polar molecular due to the linear shape of CO ₂ molecule which cancels out the polarity of the molecule. SO ₂ has an angular shape and is a polar molecule due to the electronegativity difference between the two elements in the molecule. The more polar the molecule the higher the solubility in polar solvents like water is likely to be.																	
6a(i)		Butan-2-ol is a secondary alcohol as it has 2 carbons directly attached to the carbon with the -OH group. There are three alcohol isomers of butan-2-ol: <table><tr><td> butan-1-ol  <u>Primary Alcohol</u> 1 carbon directly attached to the carbon with the -OH group </td><td> 2-methylpropan-1-ol  <u>Primary Alcohol</u> 1 carbon directly attached to the carbon with the -OH group </td><td> 2-methylpropan-2-ol  <u>Tertiary Alcohol</u> 3 carbons directly attached to the carbon with the -OH group </td></tr></table>		butan-1-ol  <u>Primary Alcohol</u> 1 carbon directly attached to the carbon with the -OH group	2-methylpropan-1-ol  <u>Primary Alcohol</u> 1 carbon directly attached to the carbon with the -OH group	2-methylpropan-2-ol  <u>Tertiary Alcohol</u> 3 carbons directly attached to the carbon with the -OH group													
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6a(ii) Part A	Alcohols are flammable	A water bath should be used to heat the reactants instead of a Bunsen burner due to the flammability of the chemicals being heated.																	
6a(ii) Part B	orange → green	<table><tr><th>Oxidising Agent</th><th>Start Colour</th><th>End Colour</th></tr><tr><td>Acidified Dichromate</td><td>Orange</td><td>Green</td></tr><tr><td>Benedict's/Fehling's</td><td>Blue</td><td>Brick Red (orange)</td></tr><tr><td>Hot copper (II) oxide</td><td>Black</td><td>Brown</td></tr><tr><td>Tollen's Reagent</td><td>(Colourless)</td><td>Silver mirror produced</td></tr></table>		Oxidising Agent	Start Colour	End Colour	Acidified Dichromate	Orange	Green	Benedict's/Fehling's	Blue	Brick Red (orange)	Hot copper (II) oxide	Black	Brown	Tollen's Reagent	(Colourless)	Silver mirror produced	
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6a(ii) Part C	Tertiary alcohol	<table><tr><td rowspan="3">Oxidation of Alcohols</td><td>Primary alcohol</td><td>→</td><td>Aldehyde</td><td>→</td><td>Carboxylic acid</td></tr><tr><td>Secondary alcohol</td><td>→</td><td>Ketone</td><td>→</td><td>[No oxidation]</td></tr><tr><td>Tertiary alcohol</td><td>→</td><td>X</td><td>→</td><td>[No oxidation]</td></tr></table>		Oxidation of Alcohols	Primary alcohol	→	Aldehyde	→	Carboxylic acid	Secondary alcohol	→	Ketone	→	[No oxidation]	Tertiary alcohol	→	X	→	[No oxidation]
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6a(iii) Part A	butanoic acid (or 2-methylpropanoic acid)	For wet pH paper to turn red an acid would have to dissolve in the water in the pH paper. As copper(II) oxide is an oxidising agent, only a primary alcohol will oxidise to form the carboxylic acid which would turn wet pH paper red. butan-1-ol (C₄H₉OH) would oxidise to butanoic acid. 2-methylpropan-1-ol (C₄H₉OH) would oxidise to 2-methylpropanoic acid.																	

6a(iii) Part B	$\begin{array}{c} \text{C}_4\text{H}_9\text{OH} + \text{H}_2\text{O} \\ \downarrow \\ \text{C}_4\text{H}_8\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \end{array}$	Write down the main species $\text{C}_4\text{H}_9\text{OH} \longrightarrow \text{C}_4\text{H}_8\text{O}_2$																				
		Balance all atoms other than O and H (not needed in this example) $\text{C}_4\text{H}_9\text{OH} \longrightarrow \text{C}_4\text{H}_8\text{O}_2$																				
		Balance O atoms by adding H ₂ O to one side $\text{C}_4\text{H}_9\text{OH} + \text{H}_2\text{O} \longrightarrow \text{C}_4\text{H}_8\text{O}_2$																				
		Balance H atoms by adding H ⁺ to one side $\text{C}_4\text{H}_9\text{OH} + \text{H}_2\text{O} \longrightarrow \text{C}_4\text{H}_8\text{O}_2 + 4\text{H}^+$																				
		Balance charge by adding e ⁻ to the most positive side $\text{C}_4\text{H}_9\text{OH} + \text{H}_2\text{O} \longrightarrow \text{C}_4\text{H}_8\text{O}_2 + 4\text{H}^+ + 4\text{e}^-$																				
6b(i)	2-methylpentanal	2-methylpentanal -CH₃ methyl group on C₂ of main chain 5 carbons on main chain -CHO aldehyde functional group																				
6b(ii)	171°C	<table><tr><td>Aldehyde</td><td>Pentanal</td><td>Hexanal</td><td>Heptanal</td><td>Heptanal</td></tr><tr><td>Boiling Point (°C)</td><td>102</td><td>130</td><td>153</td><td>-</td></tr><tr><td>Difference</td><td></td><td>28°C</td><td>23°C</td><td>18°C</td></tr><tr><td>Prediction</td><td>-</td><td>-</td><td>-</td><td>171°C</td></tr></table>	Aldehyde	Pentanal	Hexanal	Heptanal	Heptanal	Boiling Point (°C)	102	130	153	-	Difference		28°C	23°C	18°C	Prediction	-	-	-	171°C
Aldehyde	Pentanal	Hexanal	Heptanal	Heptanal																		
Boiling Point (°C)	102	130	153	-																		
Difference		28°C	23°C	18°C																		
Prediction	-	-	-	171°C																		
6b(iii)	One from:	The longer the carbon chain the higher the boiling point The higher the number of side groups the lower the boiling point.																				
6b(iv)	Silver mirror formed	<table><tr><td>Oxidising Agent</td><td>Start Colour</td><td>End Colour</td></tr><tr><td>Acidified Dichromate</td><td>Orange</td><td>Green</td></tr><tr><td>Benedict's/Fehling's</td><td>Blue</td><td>Brick Red (orange)</td></tr><tr><td>Hot copper (II) oxide</td><td>Black</td><td>Brown</td></tr><tr><td>Tollen's Reagent</td><td>(Colourless)</td><td>Silver mirror produced</td></tr></table>	Oxidising Agent	Start Colour	End Colour	Acidified Dichromate	Orange	Green	Benedict's/Fehling's	Blue	Brick Red (orange)	Hot copper (II) oxide	Black	Brown	Tollen's Reagent	(Colourless)	Silver mirror produced					
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Tollen's Reagent	(Colourless)	Silver mirror produced																				
6c	Permanent dipole to permanent dipole attractions	The C=O bond is a polar bond due to the difference in electronegativity of the elements within the bond is large enough to form a polar covalent bond These permanent dipoles within polar covalent molecules are attracted to each other are bring the molecules closer together and raise the boiling point.																				
7a	Answer to include:	Once the contents of the beaker have been poured through the filter funnel, the beaker is rinsed with deionised water to ensure every drop of the solution is washed into the standard flask. The filter funnel this then rinsed with deionised water (inside and out). The standard flask is then filled up the mark with the last few drops being added by dropper so that the bottom of the meniscus is on the line on the flask.																				
7b(i)	Permanganate is self-indicating	Permanganate MnO ₄ ⁻ ions are purple and these ions react in the titration to become colourless Manganese Mn ²⁺ ions.																				
7b(ii)	H ⁺ ions are reactants	Acid contains H ⁺ ions which are a reactant in the reaction, appearing before the arrow.																				
7b(iii) Part A	Sample 1 is ignored as it is inaccurate	The 1 st Titre is the rough titre which is not designed to be accurate but gives an idea of the rough volume at which the colour change takes place. Subsequent titrations are carried out with the majority of the volume added in one big addition and the remaining volume added drop by drop until the colour change is accurately determined.																				
7b(iii) Part B	0.0502	no. of mol = volume x concentration = 0.01455litres x 0.02mol l ⁻¹ = 0.000291mol $\begin{array}{c} 5\text{Fe}^{2+} + 8\text{H}^+ + \text{MnO}_4^- \longrightarrow 5\text{Fe}^{3+} + \text{Mn}^{2+} + 4\text{H}_2\text{O} \\ \begin{array}{ccc} 5\text{mol} & 1\text{mol} & \\ 0.001255\text{mol} & 0.000291\text{mol} & \end{array} \end{array}$ $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.001255\text{mol}}{0.025\text{litres}} = 0.0502\text{mol l}^{-1}$																				
7b(iii) Part C	Solution whose concentration is accurately known	Standard solutions have a very accurately known concentration which can then be used to work out the concentration or quantity of another substance.																				
7b(iii) Part D	Pipette	Pipettes, when used with a pipette filler for safety reasons, is more accurate than using measuring cylinders and beakers for accurately measuring volumes of liquids.																				
7c	14.1	5 tablets = 0.00126mol Fe ∴ 1 tablet = 0.000252mol Fe 1 mole of Fe = 55.8g mass = no. of mol x gfm = 0.000252mol x 55.8g mol ⁻¹ = 0.0141g = 14.1mg																				

7d	24.3	100g breakfast cereal contains 12.0mg Fe 30g breakfast cereal contains 12.0mg Fe $\times \frac{30}{100} = 3.6\text{mg Fe}$ $\% \text{ Fe in 30g cereal} = \frac{\text{mass of Fe}}{\text{Recommended mass}} \times 100 = \frac{3.6\text{mg}}{14.8\text{mg}} \times 100 = 24.3\%$														
8	Open Question Answer to Include:	3 mark answer Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	2 mark answer Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	1 mark answer Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.												
9a(i)	Diagram showing:															
9a(ii)	hydrophobic	<table><tr><th>Part of Soap</th><th>Water</th><th>Polarity</th><th>Mechanism</th></tr><tr><td>Head</td><td>Hydrophilic</td><td>Polar</td><td>Polar head stays outside non-polar grease droplet and surrounds droplet until droplet is able to dissolve in water</td></tr><tr><td>Tail</td><td>Hydrophobic</td><td>Non-polar</td><td>Non-polar tail enters non-polar grease droplet but head cannot enter droplet.</td></tr></table>			Part of Soap	Water	Polarity	Mechanism	Head	Hydrophilic	Polar	Polar head stays outside non-polar grease droplet and surrounds droplet until droplet is able to dissolve in water	Tail	Hydrophobic	Non-polar	Non-polar tail enters non-polar grease droplet but head cannot enter droplet.
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Tail	Hydrophobic	Non-polar	Non-polar tail enters non-polar grease droplet but head cannot enter droplet.													
9b(i)	Kills Bacteria	Oxidising agents can be used kill bacteria and fungi. Oxidising agents can inactive viruses.														
9b(ii)	0.036litres or 36cm ³	$\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.051\text{g}}{34\text{g mol}^{-1}} = 0.0015\text{mol}$ $2\text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{O}_2$ $\frac{2\text{mol}}{0.0015\text{mol}}$ $\frac{1\text{mol}}{0.00075\text{mol}}$ $\text{Volume} = \text{no. of mol} \times \text{Molar Volume} = 0.00075\text{mol} \times 24 \text{ litres mol}^{-1} = 0.036 \text{ litres}$														
9c(i)	amino acids	Proteins are polymers made from amino acid monomers. The breakdown of protein into amino acids is a hydrolysis reaction.														
9c(ii) Part A	Amide Link (or peptide link)	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}$ carboxyl group + $\begin{array}{c} \text{H} \\ \\ \text{H}-\text{N}- \end{array}$ amine $\xrightarrow[\text{water removed at join}]{\text{condensation}}$ $\begin{array}{c} \text{O} \quad \text{H} \\ \parallel \quad \\ -\text{C}-\text{N}- \end{array}$ amide link														
9c(ii) Part B	One from:															
9c(iii) Part A	Denaturing	Enzymes are specifically-shaped proteins. Any change to the shape of an enzyme will prevent the enzyme from fitting the exact shape of the substrate molecule it catalyses the reaction for.														
9c(iii) Part B	Increase in temperature or change in pH	Enzymes have optimum temperatures and pH conditions which they work best at. Raising the temperature above 37°C will denature the enzyme. Altering the conditions of pH will also change the shape of the enzyme.														

9d(i)	Condensation	Condensation reactions involve the joining together of two molecules with the removal of a small molecule (usually water). When the acetic anhydride splits in half, one half joins to the N atom in ethylene diamine and water is removed between the two molecules.																												
9d(ii)	Diagram showing:																													
10a(i)	40.2	Relative concentration = $\frac{\text{Component peak area}}{\text{Total Peak Area}} \times 100 = \frac{31909}{79310} \times 100 = 40.2\%$																												
10a(ii)	geranyl acetate	<table><tr><td>Lavender Oil A Peak</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Component</td><td>1,8-cineole</td><td>Linalool</td><td>Camphor</td><td>linalyl acetate</td><td>geranyl acetate</td><td>farnesene</td></tr><tr><td>Retention Time (min)</td><td>17</td><td>18</td><td>19</td><td>20.75</td><td>21.25</td><td>24</td></tr><tr><td>Lavender Oil B Peak</td><td>1st</td><td>2nd</td><td>3rd</td><td>4th</td><td>No peak</td><td>5th</td></tr></table>	Lavender Oil A Peak	1	2	3	4	5	6	Component	1,8-cineole	Linalool	Camphor	linalyl acetate	geranyl acetate	farnesene	Retention Time (min)	17	18	19	20.75	21.25	24	Lavender Oil B Peak	1 st	2 nd	3 rd	4 th	No peak	5 th
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10b	£0.0272 or 2.72p	<table><tr><td>1cm³ mouthwash</td><td>contains</td><td>0.92mg 1,8-cineole</td></tr><tr><td>500cm³ mouthwash</td><td>contains</td><td>0.92mg × ⁵⁰⁰/₁</td></tr><tr><td></td><td></td><td>= 460mg 1,8-cineole</td></tr><tr><td></td><td></td><td>= 0.46g 1,8-cineole</td></tr><tr><td>1000g 1,8-cineole</td><td>costs</td><td>£59.10</td></tr><tr><td>0.46g 1,8-cineole</td><td>costs</td><td>£59.10 × ^{0.46g}/_{1000g}</td></tr><tr><td></td><td></td><td>= £0.0272 = 2.72p</td></tr></table>	1cm ³ mouthwash	contains	0.92mg 1,8-cineole	500cm ³ mouthwash	contains	0.92mg × ⁵⁰⁰ / ₁			= 460mg 1,8-cineole			= 0.46g 1,8-cineole	1000g 1,8-cineole	costs	£59.10	0.46g 1,8-cineole	costs	£59.10 × ^{0.46g} / _{1000g}			= £0.0272 = 2.72p							
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		= £0.0272 = 2.72p																												
10c(i)	Diagram showing:																													
10c(ii)	C ₁₅ H ₂₄	One isoprene unit = C₅H₈ ∴ three isoprene units = C₁₅H₂₄ 2-methylbuta-1,3-diene																												

2017 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	81	14	2	3	0	0
2	D	2	6	6	86	0	0
3	C	9	6	76	9	0	0
4	B	10	86	3	1	0	0
5	D	6	7	5	82	0	0
6	C	0	2	84	4	0	0
7	B	1	95	3	2	0	0
8	C	18	4	65	13	0	0
9	A	98	1	1	1	0	0
10	C	6	11	68	16	0	0
11	B	17	61	7	15	0	0
12	C	3	3	85	9	0	0
13	B	25	67	4	4	0	0
14	B	18	79	3	0	0	0
15	C	4	9	84	3	0	0
16	A	83	2	2	12	0	0
17	D	1	7	3	89	0	0
18	A	76	14	5	5	0	0
19	D	4	0	2	93	0	0
20	B	22	0	2	6	0	0